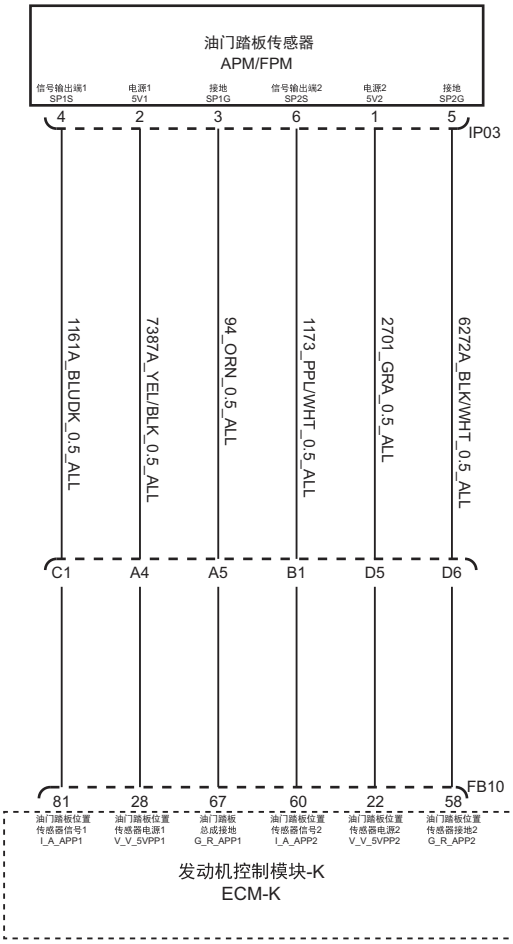
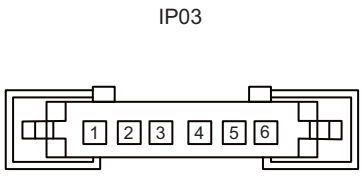
P0653

Description of fault diagnosis codes
P0653-12：Error Sensor supplies 2
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Harness or plug (short circuit of sensor circuit) • Accelerator pedal sensor • Common rail pressure sensor • Inlet air temperature pressure sensor • Camshaft position sensor
Description of circuit diagram
The engine control module will apply 5V voltage to some sensors that are divided into two groups by the system respectively. Relatively, in case of short circuit on power supply of one sensor, the sensor in the same group may be subject to short circuit fault at the same time.

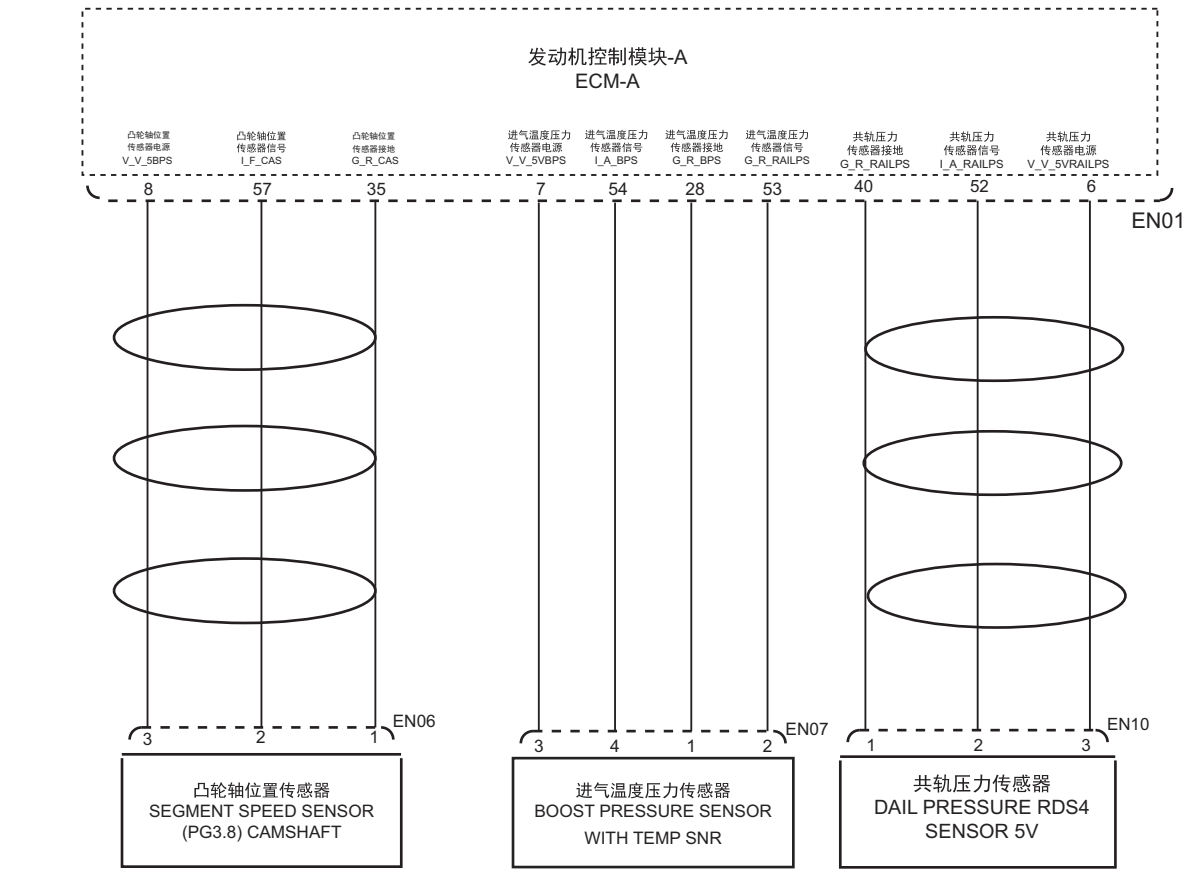
Circuit diagram

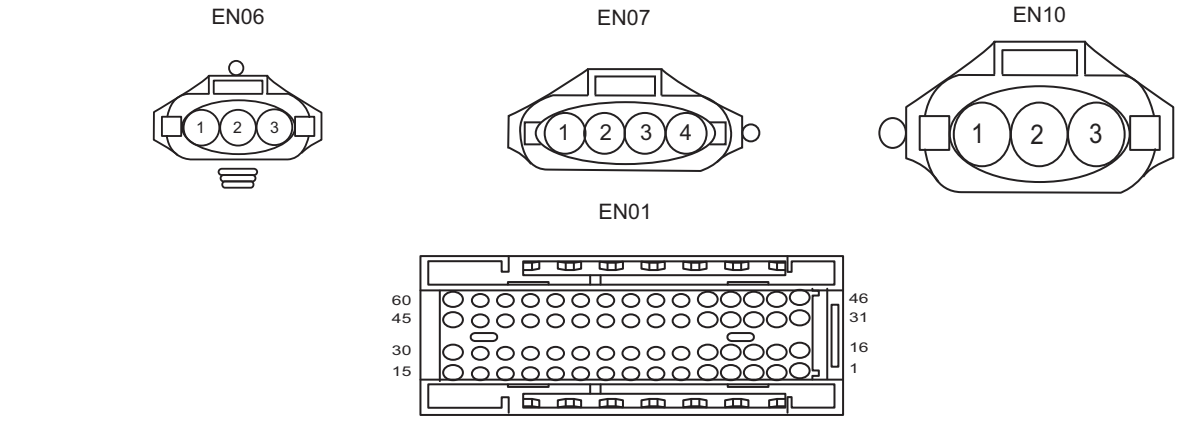






SV610800


发动机控制模块-A
ECM-A
凸轮轴位置传感器电源 V_V_5BPS
凸轮轴位置传感器信号 I_F_CAS
凸轮轴位置传感器接地 G_R_CAS
进气温度压力传感器电源 V_V_5VBPS
进气温度压力传感器信号 I_A_BPS
进气温度压力传感器接地 G_R_BPS
进气温度压力传感器信号 G_R_RAILPS
共轨压力传感器接地 G_R_RAILPS
共轨压力传感器信号 I_A_RAILPS
共轨压力传感器电源 V_V_5VRAILPS
8 57 35 7 54 28 53 40 52 6
EN01
EN06
EN07
EN10
凸轮轴位置传感器
SEGMENT SPEED SENSOR
(PG3.8) CAMSHAFT
进气温度压力传感器
BOOST PRESSURE SENSOR
WITH TEMP SNR
共轨压力传感器
DAIL PRESSURE RDS4
SENSOR 5V


EN06
EN07
EN10
EN01
60 45 30 15
46 31 16 1

SV610823

Troubleshooting procedures		
Step 1	General inspection.	
		1 Check the fuse for damage. 2 Check battery capacity. 3 Check if there is any damage, bad contact, aging and looseness phenomenon on harness plug of engine control module and sensor. 4 Check whether the result is normal?
		No → Repair or replace the faulted parts.
Yes ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to diagnosis interface. 2 Start up the engine, and maintain idle running. 3 Diagnose the engine control system with diagnostic apparatus. 4 If there is any fault code except for P0643-12?
		Yes → Refer to:DTC Summary List (EU5).
No ↓		
Step 3	Check the power supply circuit of sensor.	
		1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of accelerator pedal sensor. 3 Disconnect the harness plug of common rail pressure sensor. 4 Disconnect the inlet air temperature pressure harness plug. 5 Disconnect the harness plug of camshaft position sensor. 6 Turn the ignition switch to ON position. 7 Measure the voltage between the harness plug of sensor and ground. Measure circuit: voltage between terminal 2 of IP03 plug and ground. Measuring circuit: voltage between terminal 3 of plug EN10 and ground. Measuring circuit: voltage between terminal 3 of plug EN07 and ground. Measuring circuit: the voltage between terminal 1 of plug EN06 and ground. **Standard value: 4.8~5.2V** 8 Check whether the result is normal?

		Yes ➞ Go to Step 6.
No ↓		
Step 4	Check the sensor circuit (open circuit).	
		1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of engine control module. 3 Disconnect the harness plug of accelerator pedal sensor. 4 Disconnect the harness plug of common rail pressure sensor. 5 Disconnect the inlet air temperature pressure harness plug. 6 Disconnect the harness plug of camshaft position sensor. 7 Measure the resistance between harness plug of engine control module and the sensor. Measuring circuit: resistance between terminal 2 on plug IP03 and terminal 28 of plug FB10. Measuring circuit: the resistance between terminal 3 of plug EN10 and terminal 6 of plug EN01. Measuring circuit: the resistance between terminal 3 of plug EN07 and terminal 7 of plug EN01. Measuring circuit: the resistance between terminal 1 of plug EN06 and terminal 8 of plug EN01. **Standard value: ≈ 0Ω** 8 Check whether the result is normal?
		No ➞ Repair or replace the faulted parts.
Yes ↓		

Step 5	Check the sensor circuit (short circuit).	
		1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of engine control module. 3 Disconnect the harness plug of accelerator pedal sensor. 4 Disconnect the harness plug of common rail pressure sensor. 5 Disconnect the inlet air temperature pressure harness plug. 6 Disconnect the harness plug of camshaft position sensor. 7 Measure the voltage between harness plug of engine control module and ground. Measuring circuit: the voltage between terminal 28 of plug FB10 and ground. Measuring circuit: the voltage between terminal 6 of plug EN01 and ground. Measuring circuit: the voltage between terminal 7 of plug EN01 and ground. Measuring circuit: the voltage between terminal 8 of plug EN01 and ground. **Standard value: ≈ 0V** 8 Measure the resistance between the harness plug of sensor and ground. Measuring circuit: the resistance between terminal 2 of plug IP03 and ground. Measuring circuit: the resistance between terminal 3 of plug EN10 and ground. Measuring circuit: the resistance between terminal 3 of plug EN07 and ground. Measuring circuit: the resistance between terminal 1 of plug EN06 and ground. **Standard value: ∞** 9 Check whether the result is normal?
		No → Repair or replace the faulted parts.
Yes ↓		
Step 6	Check the sensor.	
		1 Replace the sensor.
Next ↓		
Step 7	Complete the check and confirm whether the system works normally nor not.	